Curriculum Vitae

Lukas William Mayer

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Education

Ph.D. Cognitive Sciences
Thesis: TBD
Advisor: Professor Mark Steyvers

University of California, Irvine
Irvine, California, USA
2023 - 2028 (Expected)

M.Sc. Behavioral and Data Science (Distinction)
Thesis: Cognition-Inspired Heuristics Improve Machine-Learning Predictions in Multi-Attribute Multi-Alternative Choice
Advisor: Professor Nick Chater

University of Warwick
Coventry, England, UK
2021 - 2022

B.Sc. with Honors Psychology (First-Class)
Thesis: Cooperation does not improve selective and/or sustained attention
Advisors: Dr. John Michael and Dr. Ross Goutcher

University of Stirling
Stirling, Scotland, UK
2017 - 2021

Research Interests

I am broadly interested in investigating how people use AI tools through a Cognitive Science lens, placing my work at the intersection of Computational Cognitive Science and Human-Computer Interaction. In recent projects, I have been leveraging Bayesian cognitive models to understand how people weigh up the attributes of these novel systems, and learn to navigate their trade-offs. By committing to formal, mechanistic models of human behavior, my research aims to advance human-centric AI in a manner that both improves our theoretical understanding of cognition, while also offering practical avenues to address real-world problems.

Keywords: *Bayesian Cognitive Modelling, Human-AI Collaboration, Hybrid Intelligence, Human-centered AI, Decision-making*

Publications

In preparation

- **Mayer, L. W.** & Steyvers, M. (In preparation). Modeling advice fatigue: When people disable AI assistance

Peer-Reviewed

- **Mayer, L. W.**, Karny, S., Ayoub, J., Song, M., Tian, D., Moradi-Pari, E., & Steyvers, M. (2026). Human–AI collaboration: trade-offs between performance and preferences. Cognitive Research: Principles and Implications, 11(1), 18.
- Steyvers, M., Tejeda, H., Kumar, A., Belem, C., Karny, S., Hu, X., **Mayer, L. W.** & Smyth, P. (2025). What large language models know and what people think they know. Nature Machine Intelligence, 7(2), 221-231.
- Bocheva, D., Hinds, J., Brown, O., & **Mayer, L. W.** (2025). Examining membership across multiple online communities in the "manosphere". New Media & Society, 14614448251388648.
- **Mayer, L. W.**, Bocheva, D., Hinds, J., Brown, O., Piwek, L., & Ellis, D. A. (2025). Optimeet: A computational tool to enhance participant attendance in group research. Behavior Research Methods, 57(10), 289.

Pre-prints

- Steyvers, M., & **Mayer, L. W.** (2025). When not to help: planning for lasting human-AI collaboration. arXiv preprint arXiv:2508.01837.

Presentations & Posters

- **Mayer, L. W.** (2026). *Please Don't Turn Me Off: Optimizing AI Advice Timing for Human-AI Collaboration*. 59th Annual Meeting of the Society for Mathematical Psychology (MathPsych), Montreal, Quebec, Canada.
- **Mayer, L. W.** (2026). *When to shut up: Using deliberately annoying AI assistance to enhance human-AI collaboration*. Edwards Bayesian Research Conference, California State University Fullerton, Fullerton, CA, USA.
- **Mayer, L. W.** (2025). *Walking the line: Balancing AI advice and user frustration for robust human-AI complementarity*. NSF AI Institute for Societal Decision Making, Carnegie Mellon University, Pittsburgh, PA, USA.
- Steyvers, M., & **Mayer, L. W.** (2025). A Cognitive Framework for Timely AI Communication. In Proceedings of the Annual Meeting of the Cognitive Science Society (Vol. 47).
- Karny, S., **Mayer, L.W.**, Ayoub, J., Song, M., Su, H., Tian, D., Moradi-Pari, E., & Steyvers, M. (2024). Learning with AI Assistance: A Path to Better Task Performance or Dependence? Proceedings of the ACM Collective Intelligence Conference.

Research Experience

Academic Collaborator – *Honda Research Institute* **Honda Research Institute USA**
Principal Investigator: Professor Mark Steyvers
Key Activities: Dissemination of web-based research experiments

investigating human preferences for collaborative AI agent behaviors.

Ann Arbor, MI, USA
2023 - 2025

Research Assistant – *Center for Research and Evidence on Security Threats*
Principal Investigator: Professor Adam Joinson
Key Activities: Conducted large-scale mixed methods study on predicting online group communications translating to real-world actions; developed algorithms for complex research appointment scheduling.

University of Bath
Bath, England, UK
2022 - 2023

Research Assistant and Summer Bursary Student – *Center for Cognition in Complex Environments*
Principal Investigator: Dr. Ailsa E. Millen
Key Activities: Execution and data analysis of experiments concerning face matching and concealed identity recognition.

University of Stirling
Stirling, Scotland, UK
2019 - 2021

Research Placement Student – *Center for Cognition in Complex Environments*
Principal Investigator: Dr. Jamie Murray
Key Activities: Automated video analysis for discrimination learning in lumpfish.

University of Stirling
Stirling, Scotland, UK
2021

Awards & Honors

- Psychonomic Society 2026 **Best Article** for Cognitive Research: Principles and Implications - **$1.000**
- Center for Theoretical Behavioral Sciences Luce Family **Summer Fellowship** (2026) - **$10.000**
- Center for Theoretical Behavioral Sciences 2026 **Conference Support Award** - **$1.500**
- Center for Theoretical Behavioral Sciences 2025 **Conference Support Award** - **$1.500**
- **Best Overall Score** in Behavioral and Data Science (2022)
- **Peter McEwan Prize** for the Best Single Honors Psychology Student (2021)
- **Alan Baddeley Prize** for the Best Psychology Dissertation (2021)
- **My Stirling Award GOLD** (2021) in recognition of an exceptional depth and breadth of extracurricular activities
- Honorable Mention for **Best Placement Student** (2021)

Teaching Experience

- **Peer-to-Peer Tutor in Research Methods** (2019)
  - Led seminar lectures on the fundamentals of statistical analysis to groups of approximately twenty 2nd year undergraduates

- Instructed students how to navigate statistical software and troubleshoot exercises

Professional Service

Peer-Review

- Nature Communications (1)
- Psychological Review (2)
- Judgment and Decision Making (1)

Organizations

- Cognitive Science Society
- Society for Judgment and Decision Making
- Society for Mathematical Psychology

Professional Development

- **Mentoring Excellence Certificate** (2026), University of California, Irvine, Irvine, CA, USA
- **Digital Methods Initiative Winter School: The use and misuse of Open Source Intelligence** (2023), University of Amsterdam, Amsterdam, North Holland, The Netherlands
- **International Interdisciplinary Computational Cognitive Science Summer School** (2023), University of Tübingen, Tübingen, Baden-Württemberg, Germany
- **NeuroMatch Academy Summer School: Deep Learning** (2022), NeuroMatch Academy, Los Angeles, California, USA

Extracurricular Activities

- Faculty Officer: Faculty of Natural Sciences (2020-2021)
- Secretary: Powerlifting (2019-2021)
- Faculty Officer: Department of Psychology (2019-2020)
- Module representative: Brain and Behavior (2018), Research Methods (2018), Clinical and Health Psychology (2019), Developmental Psychology (2019), Individual Differences (2019), Psychology Research Project (2019)
- Team Captain: Powerlifting (2017-2019)

Skills

- **Programming Languages**: Julia, Turing.jl, GDscript, R, Python
- **Software**: Quarto (Dashboards, Websites), Godot (Game-engine), Firebase (NoSQL Database), RevealJS (Presentations), LaTeX (Markup), Typst (Markup), Org-mode (Emacs), Git (Version-control)
- **Languages**: German (Native), English (Proficient), French (Intermediate), Dutch (Beginner)

References

Available upon request.